TEARDOWN STUDY · AALTO UNIVERSITY · ELEC-E8716 ADVANCED PACKAGING

Samsung Exynos 990 Teardown

X-ray, optical microscopy, SEM and EDX analysis of a Samsung Galaxy S20 processor — reverse-engineering a real package-on-package architecture down to its interconnects and reliability weak points.

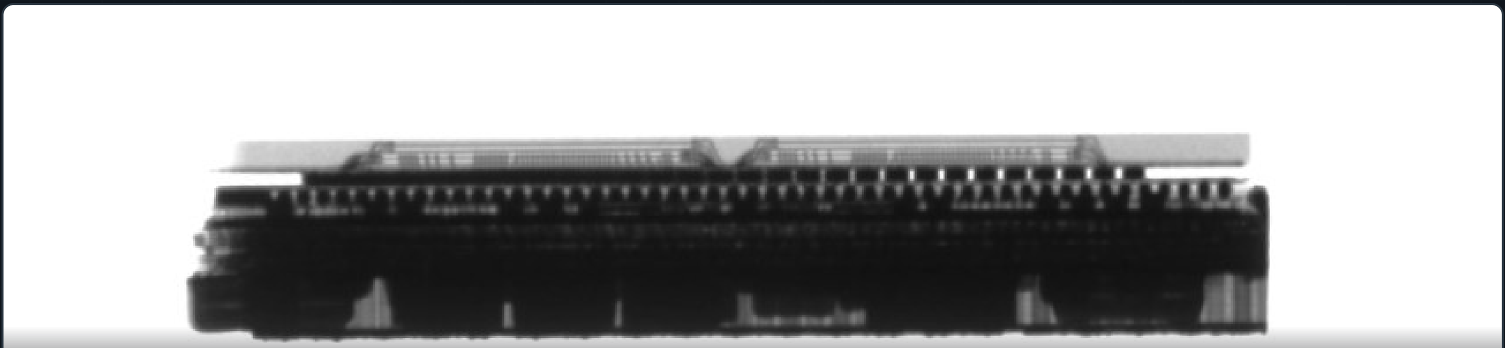

FIG.01 — Vertical X-ray cross-section of the Exynos 990 package

4
IMAGING TECHNIQUES USED

3
STACKED PACKAGE LAYERS

4
INTERCONNECT TECHNOLOGIES

0
HEAT SINKS FOUND ON THE DIE

01 WHY TEAR IT DOWN

As transistor scaling reaches its physical limits, manufacturers lean harder on **advanced packaging** — stacking and interconnecting separate dies instead of shrinking one. To understand how that actually gets built, our team physically tore down a **Samsung Galaxy S20's Exynos 990** application processor: diced it out, cross-sectioned it, and imaged the cut with four complementary techniques.

The Exynos 990 is a consumer mobile chip, not a high-reliability part — so it's a good subject for asking what packaging engineers actually choose when cost and manufacturability outweigh maximum reliability, and where those choices leave reliability risk on the table.

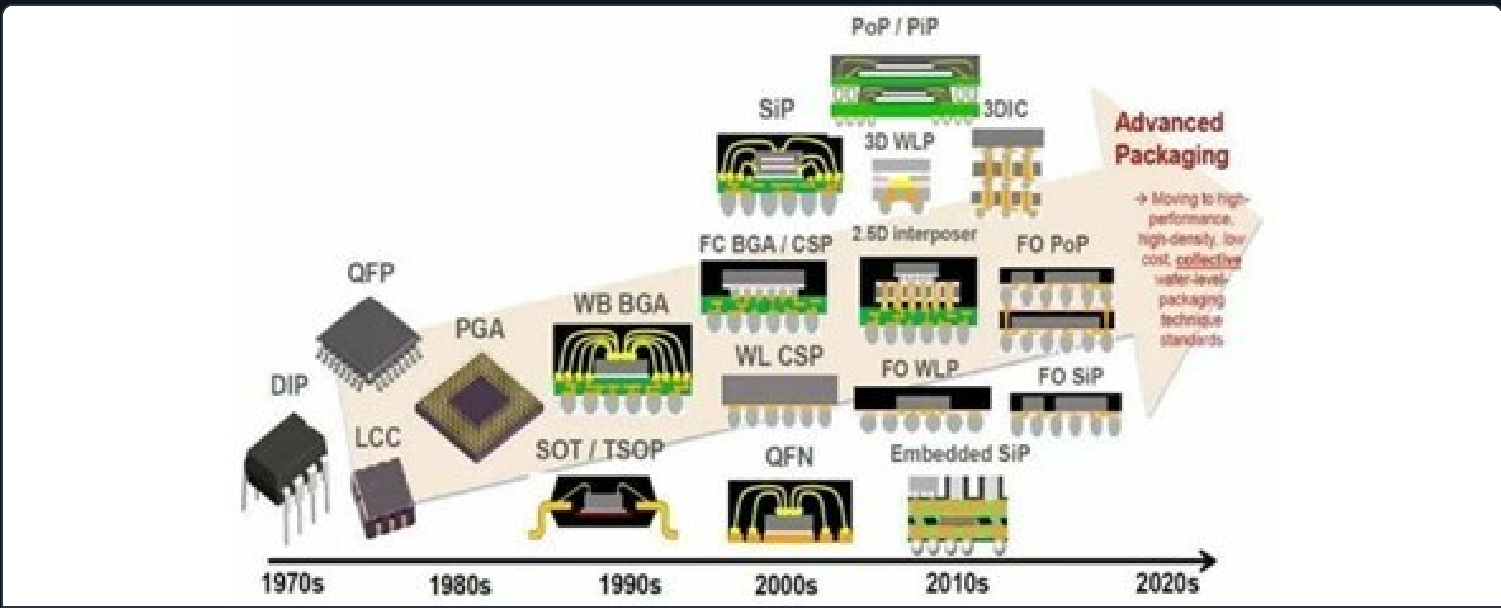

FIG.02 — Packaging evolution over the decades, DIP to modern 3D/PoP [Yole Développement, 2020]

X-RAY

Overview imaging

Non-destructive full-package view — locates the layers and solder bump rows before cutting anything.

OPTICAL

Cross-section microscopy

After grinding & polishing a cross-section, measures layer heights and interposer structure at micron scale.

SEM

Electron microscopy

Much higher resolution than optical — resolves individual bump geometry, IMC layers and wire bond profiles.

EDX

Elemental mapping

X-ray spectroscopy identifying which elements (Sn, Ag, Cu, Ni, Au...) sit where in each interconnect.

02 PACKAGE ARCHITECTURE

A three-layer package-on-package

The X-ray and optical cross-sections show a classic **package-on-package (PoP)** structure: two 4-layer DRAM stacks sit directly above the application processor, connected through a redistribution layer, with the whole assembly stacked vertically to keep the board footprint small.

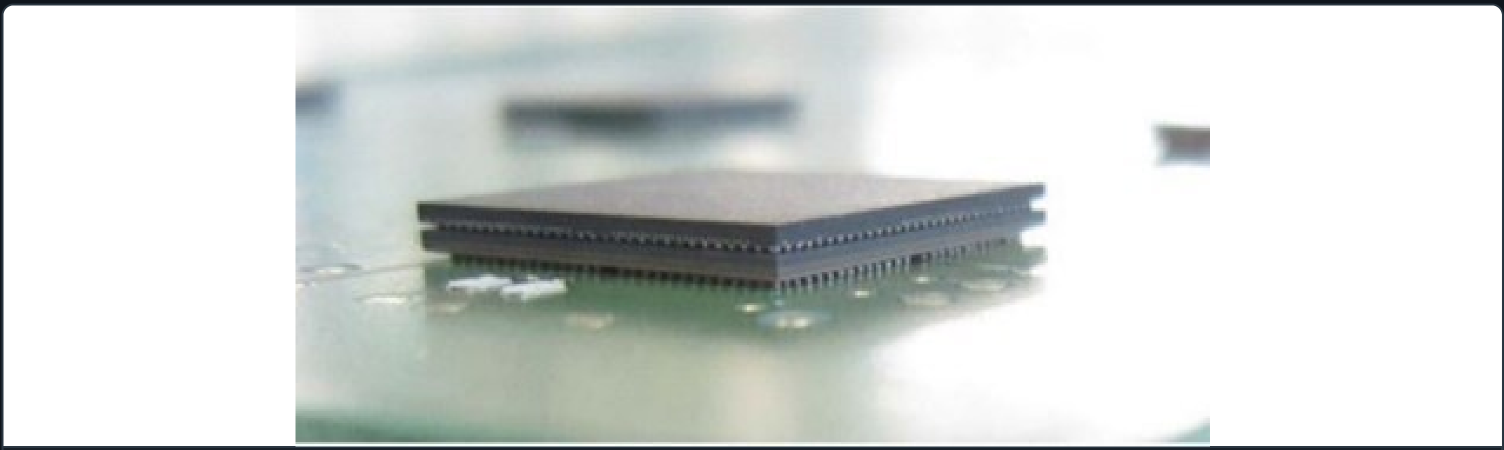

FIG.03 — Generic PoP architecture: memory vertically stacked on the processor [4]

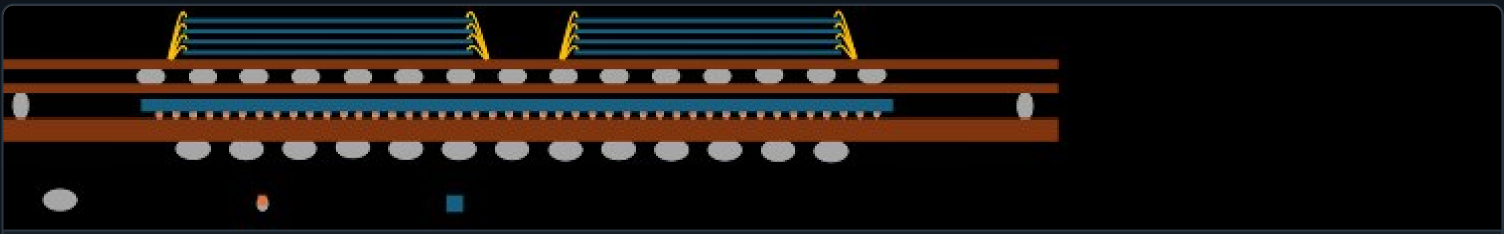

FIG.04 — Our schematic of the Exynos 990 layer stack, as reconstructed from the cross-section

470 µm

Memory stack + interposer

80 µm

Middle interposer (RDL)

105 µm

Silicon chip + metal layer

200 µm

Bottom interposer (substrate)

03 DIE ATTACH

Two adhesives, one epoxy base

Two separate die-attach joints hold the package together: the processor die to the PCB, and the stacked memory dies to each other. Both use an organic epoxy — but EDX turned up a composition difference that says something about how each was engineered.

Processor die attach

Carbon and oxygen only, with **no metallic filler** (no silver or gold particles). That keeps cost down, but also means the adhesive likely has limited thermal and electrical conductivity compared to silver-sintered alternatives.

Memory die attach

Same carbon/oxygen epoxy base, but with **traces of fluorine** — consistent with a fluorinated additive. Fluorination typically improves thermal stability and can make the adhesive more electrically insulating, which fits the tighter stack of four DRAM dies.

Both choices track with the product: a phone processor is expected to sit in roughly a **60–70°C** operating range for a few years, not the extreme thermal/mechanical cycling budget of industrial or automotive electronics — so there was little reason to pay for a premium, metal-filled die attach.

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04 INTERCONNECTS

Four interconnect technologies, one package

Every electrical connection between layers uses a different interconnect technology, sized for the job it does — from the tiny C2 copper-pillar bumps under the processor die itself, to the taller, more compliant elongated bump that has to absorb thermal mismatch between the substrate and the RDL.

C4 · MEMORY ↔ RDL

Top-layer C4 solder bump

Sn-Ag-Cu solder, Ni diffusion barrier

W 330µm · P 130 µm

150 µm

ELONGATED · RDL ↔ SUBSTRATE

Elongated solder bump

Absorbs CTE-mismatch thermal stress

W 120µm · P not measurable[1]

190 µm

C4 · SUBSTRATE ↔ PCB

Bottom-layer C4 solder bump

Sn-Cu with Al, no Ni/Ag present

W 340µm · P 210 µm

110 µm

C2 · SUBSTRATE ↔ CHIP

Copper-pillar solder bump

No Ni barrier detected

W 250µm · P 133.6 μm^2

60 µm

WIRE BOND · MEMORY STACK

Au ball-wedge wire bond

16 wire bonds, low-loop profile

W 20µm · P —

40 µm

Bump / bond height (µm), scale relative to tallest (190 µm)

1. Pitch not measurable in this cross-sectional view — the feature lies out of plane.
2. Precise SEM/EDX measurement from the C2 bump analysis (Table 2); the coarser optical estimate was ≈250 µm.

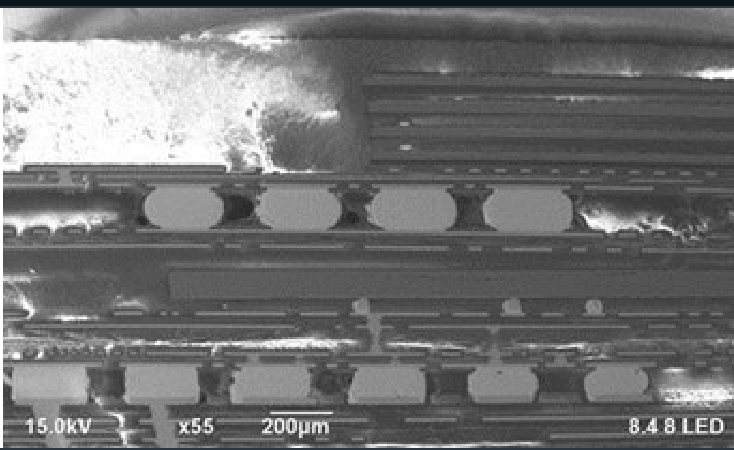

FIG.05 — SEM overview of the package, showing rows of C4 solder bumps

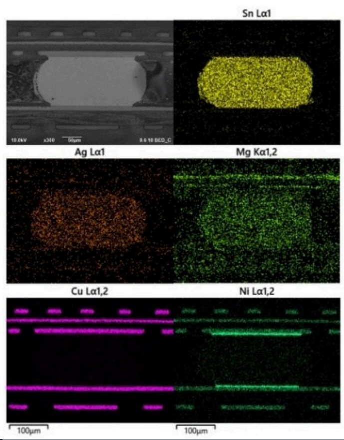

FIG.06 — C2 bump cross-section: copper pillar, solder cap, IMC layer

C4 bumps

Two different alloys in the same package: the top layer (memory → RDL) uses Sn-Ag-Cu with a nickel diffusion barrier, optimized for high-bandwidth memory connections. The bottom layer (processor → PCB) drops the Ni and Ag entirely, using a simpler Sn-Cu-Al mix suited to mechanical PCB attachment rather than signal density.

Wire bonds

16 gold ball-wedge wire bonds link the four stacked DRAM dies to the memory package. A **low-loop profile** (~40 μm height) lets the bonds run sideways instead of arching upward — essential for fitting inside a vertically stacked package, at the cost of extra mechanical stress at the wire neck.

05 RELIABILITY FINDINGS

What the cross-section says about reliability

A teardown can't predict a failure rate, but it can show where the design margin is thin. Four observations stood out as places a consumer-grade PoP trades reliability headroom for cost and size.

No heat sink on the die

`thermal risk`

No heat shield or heat sink was found over the application processor. Combined with the missing space for one in a PoP stack, this limits heat dissipation — a plausible contributor to reduced processor lifetime and to thermal-cycling stress on nearby solder joints.

No nickel barrier in the C2 bumps

`IMC growth`

Unlike every other interconnect in the package, the C2 bumps under the processor die showed **no visible Ni diffusion barrier**. That likely explains the unusually thick (3.67 μm) intermetallic-compound layer observed there — uncontrolled IMC growth can affect long-term electrical and mechanical reliability.

Inconsistent IMC composition

`variable bond strength`

EDX point analysis of the C2 bump's Cu-Sn intermetallic layer found the Cu:Sn ratio swinging from 74:26 to 59:41 across just four sample points. Different intermetallic phases have different mechanical and electrical properties, so this variation is a source of uneven bond strength across the die.

Die attach lacks conductive filler

`limited conductivity`

Neither die-attach adhesive contained the metallic fillers (silver, gold) that typically boost thermal and electrical conductivity. Reasonable for a cost-driven consumer part, but it's another point where thermal management was traded for price.

Findings summarized from the team's SEM/EDX analysis and discussion; see the full report for complete elemental data and point-analysis tables.

06 SUMMARY

What this project demonstrates

- Physical failure analysis: X-ray, optical microscopy, SEM and EDX interpretation
- Reading cross-sectional images into an accurate structural/material model
- Materials identification and reliability reasoning from elemental composition
- Connecting microscopic observations (IMC ratios, barrier layers) to system-level risk
- Collaborative technical reporting across a six-person team
- Translating dense lab data into a clear engineering narrative

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